



Ocean Sciences Campus

**Remote Sensing Analysis of Chlorophyll-a Variability and
Bathymetry in the California Current System**

Students: Aphile Vezi and Abulele Monki

Lecturers: Prof Marcello Vichi and Dr Magata Mangatane

1. Introduction

The world ocean exhibits remarkable spatial and temporal variability in phytoplankton biomass, driven by the interplay between physical oceanographic processes and nutrient availability (Chavez and Messié, 2009). Eastern boundary upwelling systems represent some of the most productive marine environments on Earth, where wind-driven coastal upwelling delivers nutrient-rich subsurface waters to the sunlit surface layer, fuelling intense phytoplankton growth detectable from space (Gruber, et al., 2011).

The California Current System (CCS) is one such region where these dynamics are particularly well expressed (Kahru, et al., 2009). The California Current System (CCS) is an eastern boundary upwelling system that extends along the Pacific coast of North America (Chavez and Messié, 2009). We selected this region for this study due to a number of factors that make the CCS an excellent area to conduct research on because of biological productivity, Equatorward winds lead to offshore Ekman transport which results in divergence along the coast and upwelling of cold, nutrient-rich water masses (Kahru, et al., 2009). As a result, there are intensive phytoplankton blooms observed in very high chlorophyll-a values at the edge of the shelf, this allows to investigate the relationship between physical forcing and biological activity (Gruber, et al., 2011). Upwelling activity varies seasonally; it is high in spring/summer and low in autumn/winter, resulting in large-amplitude variability of the chlorophyll cycle (Chavez and Messié, 2009). Lastly, the width of the continental shelf off Central California is rather small (30-50 km) but increases south of Point Conception, the shelf break and submarine canyons affect the shape of the upwelling front and productivity patterns. Also in terms of satellite observations, the CCS is one of the most well-studied regions of the ocean, with chlorophyll data available since 2002 (Gruber, et al., 2011).

2. Datasets and the listings of the metadata

2.1. Bathymetry

Bathymetric data for the California Current System was acquired from the Global Multi-Resolution Topography (GMRT) synthesis through GMRT MapTool service. The data is available in NetCDF grid format (.grd) and contains high-resolution seafloor topography that has been created by combining multibeam sonar-derived topography with regional bathymetric compilations and global elevation datasets. The bathymetry data consists of gridded elevation data (z) with a resolution of metres and a value of zero or above representing ocean elevation and topographic elevations relative to the sea level, with negative values representing ocean depth. Nesting the data was done in Python, using the xarray package for visualization and mapping.

NetCDF metadata:

- **Title:** GMRT Grid
- **Source:** GMRT MapTool Service

- **Format:** NetCDF grid (.grd)
- **Variables:**
 - ❖ z: bathymetric elevation/depth values (m)
 - ❖ x_range: longitude range
 - ❖ y_range: latitude range
 - ❖ spacing: grid spacing
- **Dimensions:** (side: 2, xysize: 256022)
- **Projection:** Cylindrical Equidistant
- **Study region extracted:** 30°N–45°N, 135°W–115°W

2.2. Chlorophyll

The monthly climatology of surface chlorophyll-a concentration data was retrieved from the European Space Agency (ESA) Ocean Colour Climate Change Initiative (OC-CCI) Version 5.0. The dataset combines multiple satellite ocean-colour sensors to produce globally consistent chlorophyll observations. The data were made available as a monthly climatological mean product at a spatial resolution of around 4 km and in NetCDF-4 format. The surface chlorophyll-a concentration is expressed as a variable named `chlor_a` (mg m⁻³). Python was used to analyze the data, and for the California Current study area (30°N–45°N, 135°W–115°W), the data were subsetted. Through pre-processing, it was determined that the latitude coordinate in the data is ordered from north to south (descending latitude values). In order to get the study area in the region and subset, it was necessary to use modified latitude slicing.

NetCDF metadata:

- **Dataset:**
ESACCI-OC-MAPPED-CLIMATOLOGY-1M_MONTHLY_4km_PML_CHL-fv5.0.nc
- **Institution:** Plymouth Marine Laboratory
- **Conventions:** CF-1.7
- **Time dimension:** 12 monthly climatological means
- **Spatial dimensions:** lat (4320) × lon (8640)
- **Latitude range:** 89.98°N to –89.98°S
- **Longitude range:** –180° to 180°
- **Variable:** `chlor_a`
- **Units:** mg m⁻³
- **Spatial resolution:** ~4 km
- **Time coverage:** 1998–2020 climatology
- **Study region extracted:** 30°N–45°N, 135°W–115°W
- **Format:** NetCDF-4 Classic

3. Bathymetry and chlorophyll concentration plots

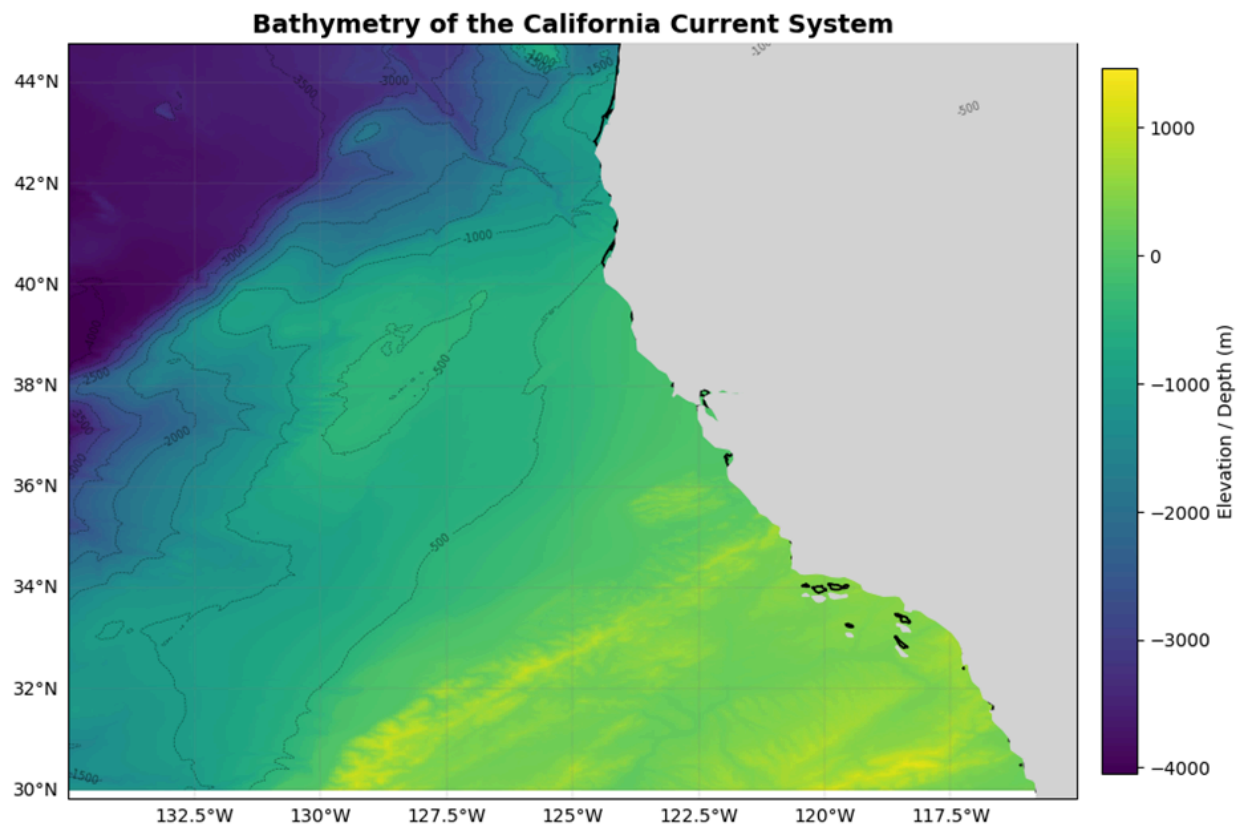


Figure 1. Bathymetry of the California Current System (30°N–44°N, 133°W–115°W) derived from the GMRT dataset. Elevation/depth is displayed in metres using the *viridis* colormap, where darker colours indicate deeper ocean regions and lighter colours represent shallower areas. Black contour lines indicate bathymetric gradients, including the continental shelf region.

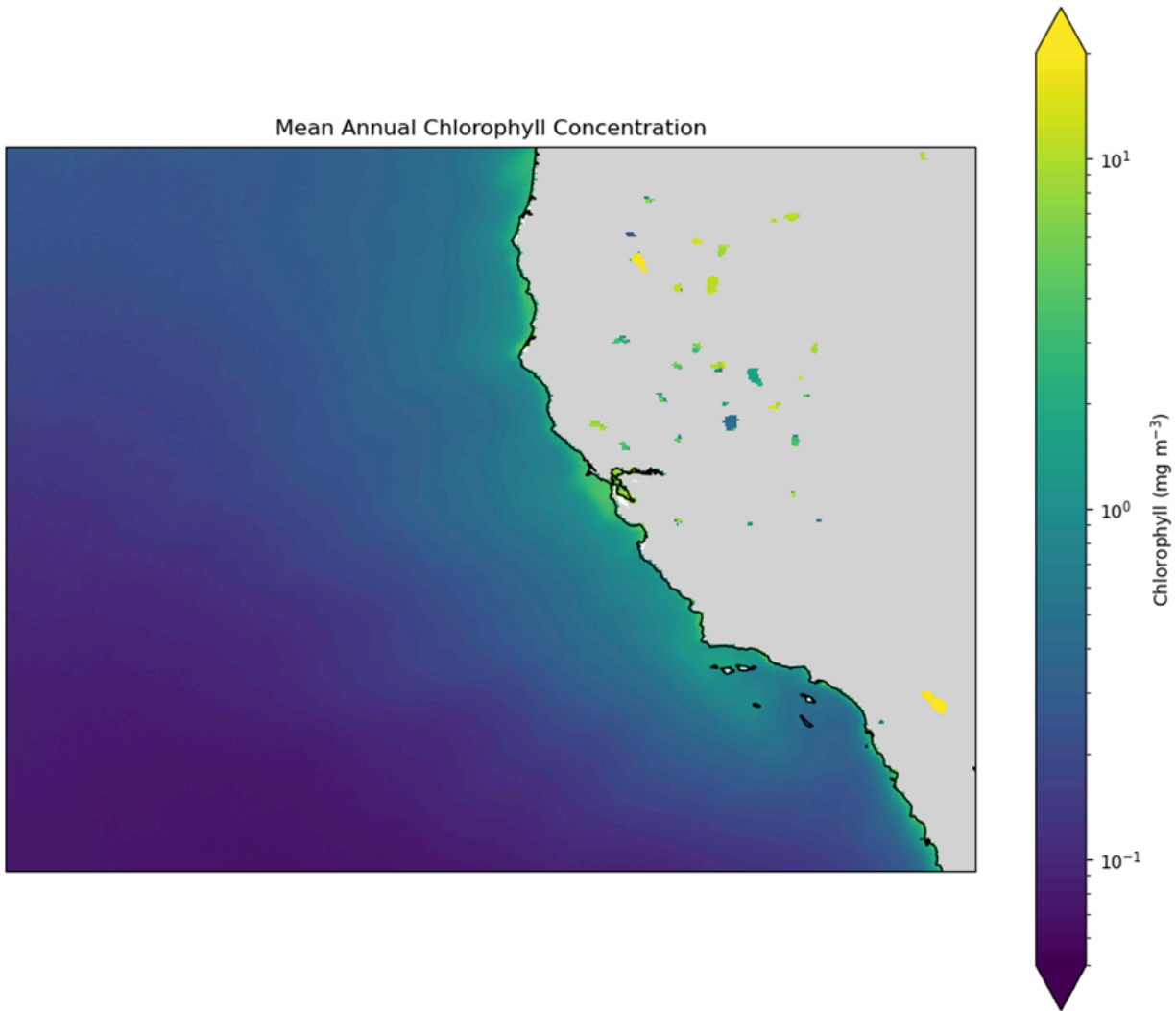


Figure 2. Mean annual surface chlorophyll-a concentration from [ESA-CCI Ocean Colour](#) , displayed using a logarithmic *viridis* colour scale ($0.1\text{--}10\text{ mg m}^{-3}$). The logarithmic scale highlights both the very low chlorophyll concentrations of the oligotrophic open ocean ($<0.1\text{ mg m}^{-3}$, shown in purple-blue colours) and the elevated concentrations in the nearshore upwelling region ($>2\text{ mg m}^{-3}$, shown in green-yellow colours). The highest chlorophyll concentrations occur close to the coastline and along the continental shelf region, reflecting enhanced biological productivity associated with coastal upwelling.

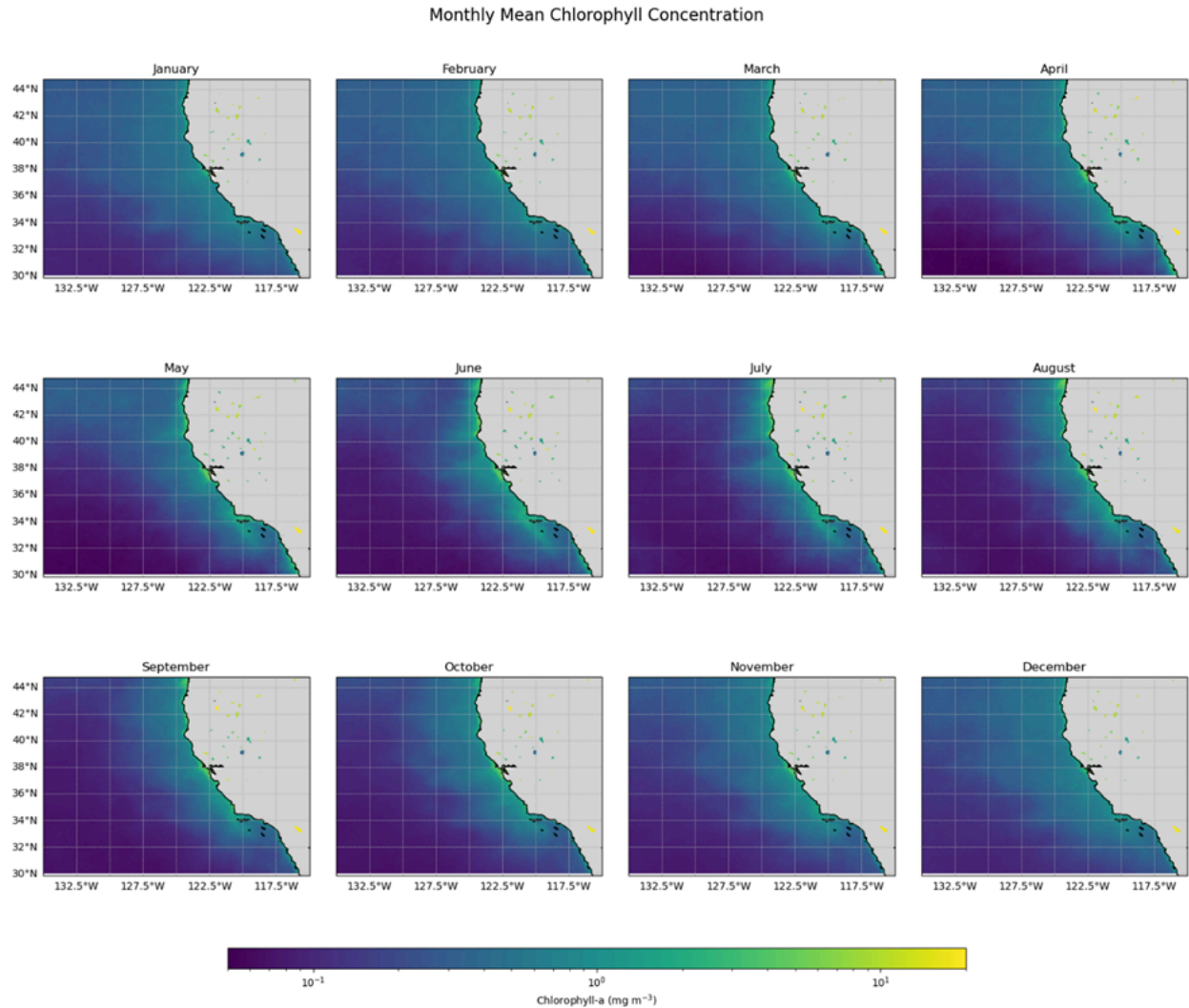


Figure 3. Twelve-panel climatological monthly mean surface chlorophyll-a concentration from [ESA-CCI Ocean Colour](#), displayed using a common logarithmic colour scale ($0.1\text{--}10 \text{ mg m}^{-3}$). There is a distinct seasonal cycle in phytoplankton biomass in the California Current System, as indicated in the figure. In the winter season (December – February) the chlorophyll levels are relatively moderate and broadly distributed near the coast, with consistently low levels in offshore waters indicative of oligotrophic open ocean conditions. From spring into early summer (March–July), the amount of chlorophyll increases considerably along the coast, especially off northern and central California, marking the start and intensification of coastal upwelling that brings nutrient-rich water to the surface ocean. Peak concentrations occur during June and July, when the coastal band of elevated chlorophyll becomes narrow, intense, and spatially continuous. Chlorophyll levels slowly decline and diminish towards the coast from late summer into autumn (August–November) with some localized elevated concentrations persist in southern California and Baja California. The low levels of chlorophyll in the offshore region throughout the year is indicative of the sharp contrast between upwelling nutrient-rich coastal waters and oligotrophic open ocean waters of the Pacific Ocean.

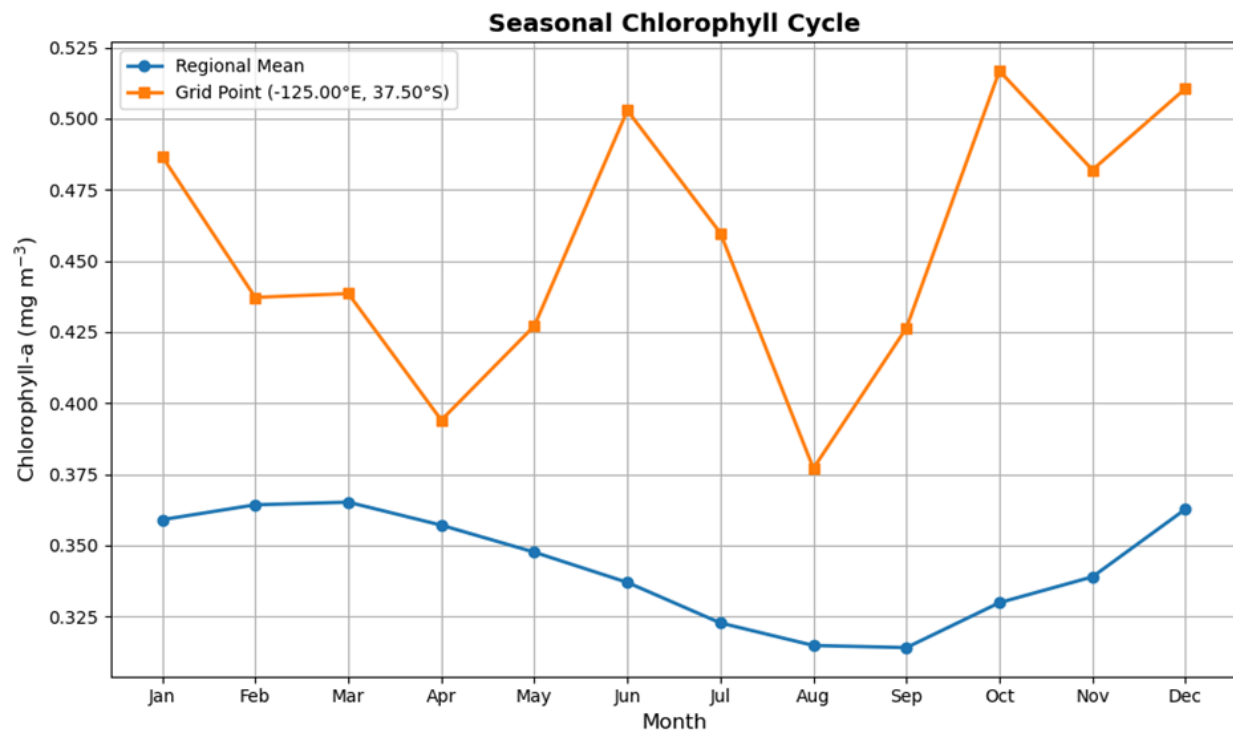


Figure 4. Seasonal cycle of surface chlorophyll-*a* concentration. The blue line represents the spatial mean across the entire study domain, while the orange line shows chlorophyll variability at a single coastal grid point (125°W, 37.5°N) located within the California upwelling region. Seasonal variation in the regional mean is relatively low, with slightly lower levels in late summer/early autumn, and higher levels in the winter months. In contrast, the coastal grid point shows much stronger seasonal fluctuations, with elevated chlorophyll concentrations during late spring, summer, and autumn, consistent with periods of enhanced coastal upwelling and nutrient supply. The peaks are observed in June, October and December whereas the minimum is observed in August. The higher variability at the coastal zones is due to the direct impact of upwelling processes closer to the continental shelf and coastal transition zone.

4. AI Usage Disclaimer

Artificial Intelligence tools were used during the preparation of this assignment as described below. All scientific interpretation, choice of region, and final review of content remained our responsibility.

How AI was used

- Code generation for visualisation (Copilot):** Copilot was primarily used to generate Python plotting scripts for the production of maps, climatologies, and seasonal chlorophyll visualisations. The datasets were uploaded and described to the AI tool in

order to generate plotting routines appropriate for the NetCDF data structure and desired figure outputs.

- **Debugging and code explanation (ChatGPT):** When errors occurred in the generated scripts, ChatGPT was used to explain error messages, identify possible causes, and suggest approaches for correcting the code. We then reviewed, modified, and implemented the corrections manually.
- **Independent implementation of basic data handling:** Dataset loading, package imports, and basic NetCDF handling procedures were completed independently by us using prior programming knowledge, without significant AI assistance.

Examples

- From this chlorophyll and bathymetry dataset, generate a seasonal time-series plot comparing the regional mean with a selected coastal grid point.
- Explain this Python error produced by the plotting code and guide me on how to fix it. E.g one of the most common error we had, is the one that says “there is no numeric data to plot”, so chatgpt helped us identify and fix that error.

5. References

- Chavez, F. P., & Messié, M. (2009). A comparison of Eastern Boundary Upwelling Ecosystems. *Progress in Oceanography*, 83(1–4), 80–96. <https://doi.org/10.1016/j.pocean.2009.07.032>
- Gruber, N., Lachkar, Z., Frenzel, H., Marchesiello, P., Münnich, M., McWilliams, J. C., Nagai, T., & Plattner, G.-K. (2011). Eddy-induced reduction of biological production in eastern boundary upwelling systems. *Nature Geoscience*, 4(11), 787–792. <https://doi.org/10.1038/ngeo1273>
- Kahru, M., Kudela, R., Manzano-Sarabia, M., & Mitchell, B. G. (2009). Trends in primary production in the California Current detected with satellite data. *Journal of Geophysical Research: Oceans*, 114(C2), C02004. <https://doi.org/10.1029/2008JC004979>